

DAY 1 - Python Basics (Theory)

What is Python?

Python is a high-level, interpreted, general-purpose programming language created by Guido van Rossum in 1991. It is simple, readable and beginner-friendly.

Features

Easy to learn; Interpreted; Platform independent; Object-oriented; Open source; Large standard library; Dynamically typed.

Applications

Web Development, Data Analysis, AI, Machine Learning, Deep Learning, Automation, Cybersecurity, Game Development, Desktop Applications, Scientific Computing.

Why Learn Python?

Easy syntax, high demand, huge community, and excellent for AI/ML.

Program

A program is a set of instructions given to a computer to perform a specific task.

Programming Language

A language used to write instructions for a computer.

Interpreter

Executes Python code one line at a time.

Compiler vs Interpreter

Compiler	Interpreter
Entire program	One line at a time
Executable created	No executable
Examples: C, C++	Examples: Python, JavaScript

Syntax

Rules for writing Python code. Errors produce SyntaxError.

Identifiers

Names for variables/functions/classes. Use letters, digits, underscores; cannot start with digit; no spaces/special characters; cannot be keywords; case-sensitive.

Keywords

Reserved words: if, else, for, while, True, False, None, class, return, import.

Variables

Named memory locations that store data.

Data Types

Type	Description	Example
int	Whole numbers	25
float	Decimal	3.14
str	Text	Python
bool	Logical	True
list	Mutable	[1,2,3]
tuple	Immutable	(1,2,3)
set	Unique	{1,2,3}
dict	Key-value	{"name": "Sneha"}

Input & Output

Input is data entered by the user. Output is the result produced.

Indentation

Python uses indentation to define code blocks.

Comments

Single-line comments use #. Triple quotes are commonly used for documentation.

Summary

Topics covered: Python, features, applications, program, programming language, interpreter, compiler vs interpreter, syntax, identifiers, keywords, variables, data types, input/output, indentation, comments.